

University of Minho
School of Engineering



Cryptography & Network Security

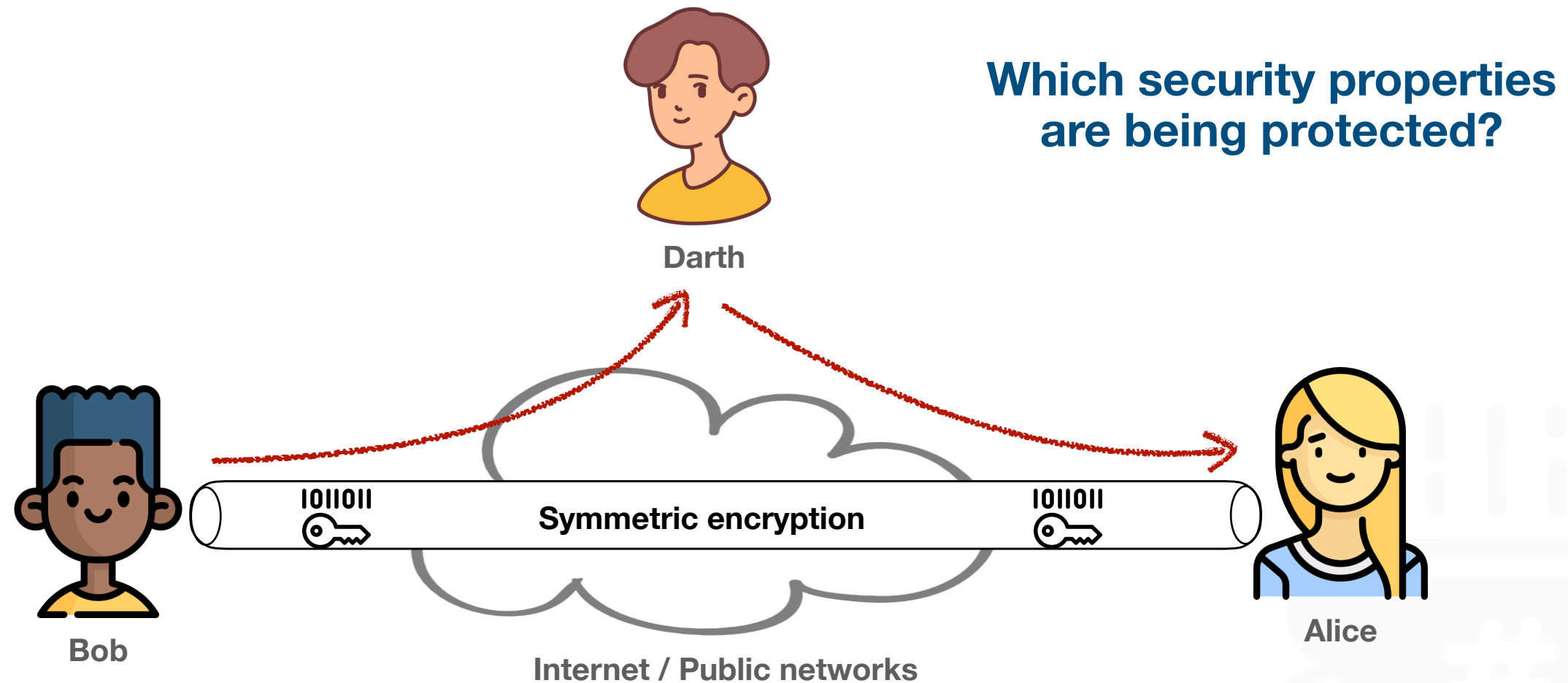
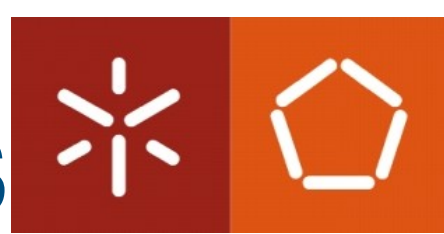
Bachelor in Telecommunications and Informatics Engineering

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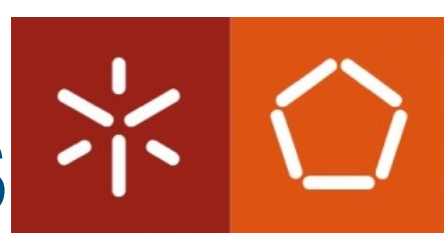
2024/2025

Cryptographic Hash Functions



- Confidentiality
- Integrity
- Availability
- Authenticity
- Non-repudiation

Cryptographic Hash Functions



Hash functions

A *hash function* H accepts a variable-length block of data M as input and produces a fixed-size hash value h .

$$h = H(M)$$

Desired properties:

- The results of applying it to a large set of inputs will produce outputs that are evenly distributed and apparently random;
- A change to any bit in M results, with high probability, in a change to h .

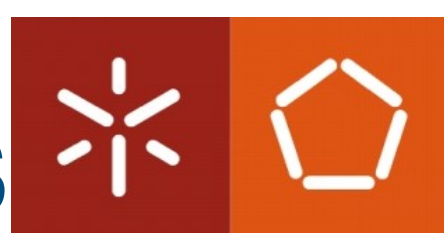
Cryptographic Hash Functions

Hash functions for which is computationally infeasible to find:

- A data object that maps to the same hash result;
- Two data objects that map to the same hash result.

Hash functions are often used to determine whether data has changed. <Integrity>

Cryptographic Hash Functions

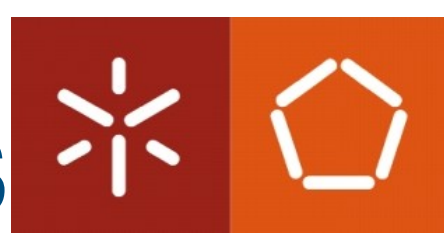


Applications of Cryptographic Hash Functions

- » Message Authentication
- » Digital Signatures
- » One-Way Password File
- » Intrusion Detection
- » Virus Detection
- » Pseudorandom Functions (PRF)
- » Pseudorandom Number Generator (PRNG)



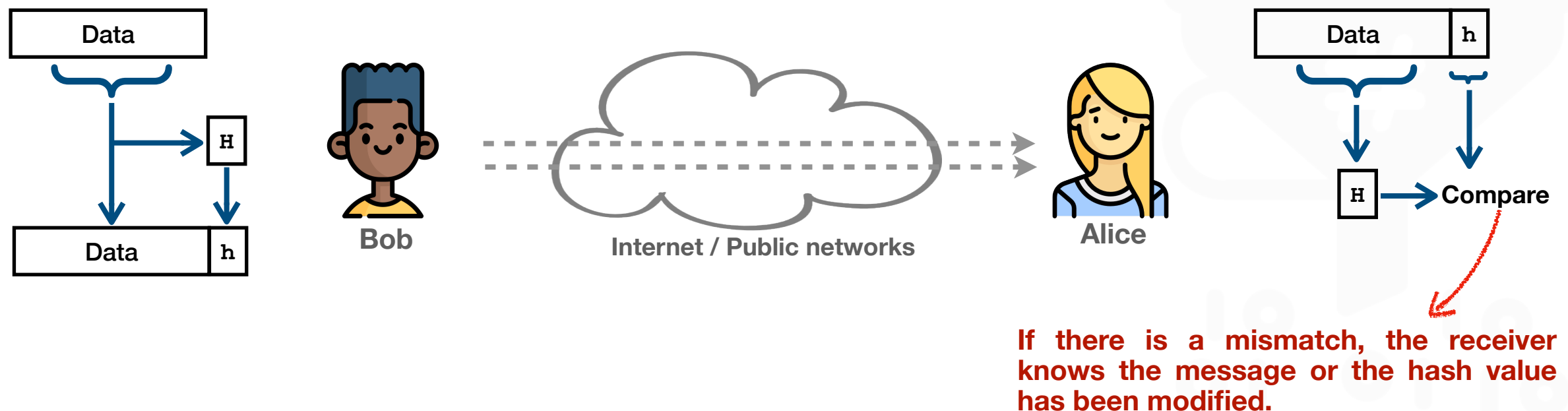
Cryptographic Hash Functions



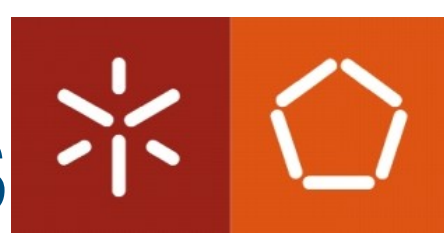
Message Authentication

A mechanism or service used to verify a message's integrity.

When used to ensure the sender's identity, the hash function value is also referred to as a *message digest*.



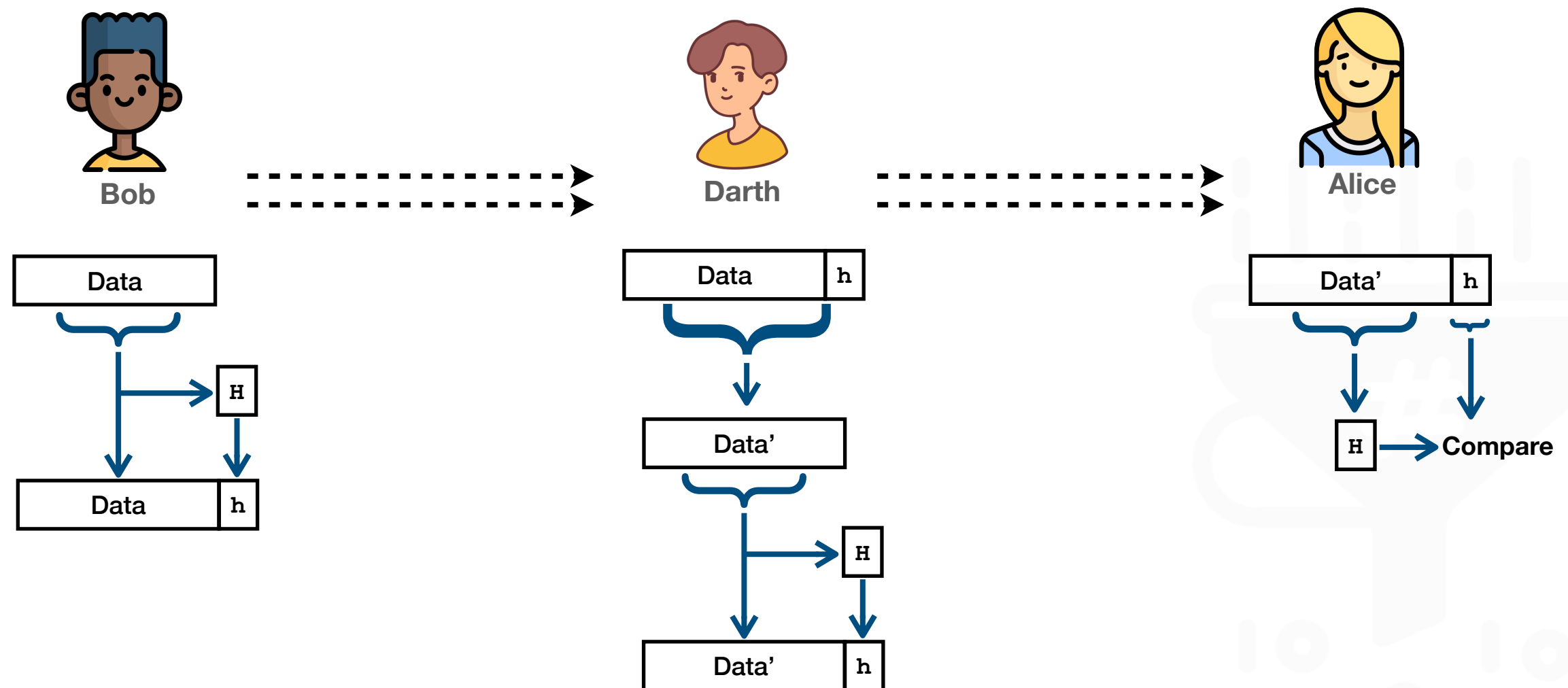
Cryptographic Hash Functions



Message Authentication

Attack against Hash Function

The hash value must be transmitted securely.



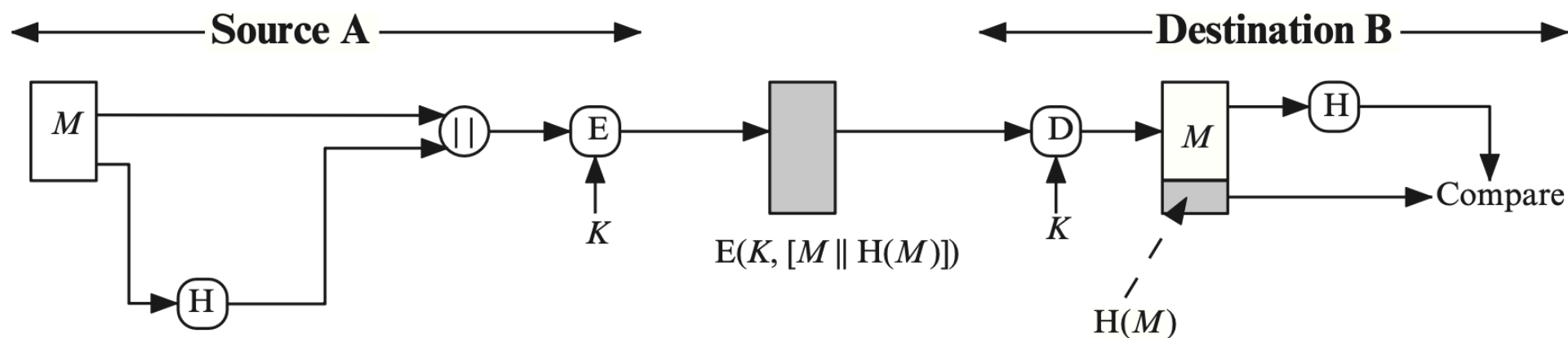
Man-in-the-middle attack

Cryptographic Hash Functions

Message Authentication



Using hash code to provide message authentication

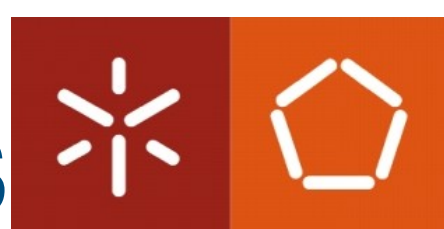


The message plus the hash code is encrypted using symmetric encryption.

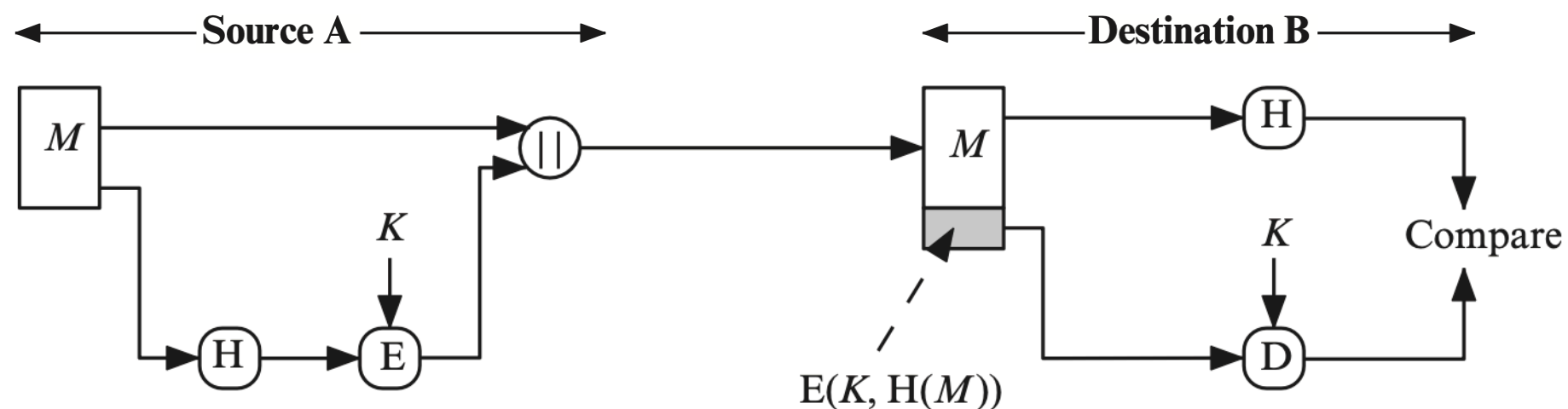
- As A and B share the key, the message must have come from A and has not been altered.
- By encrypting the entire message plus the hash code, confidentiality is also provided.

Cryptographic Hash Functions

Message Authentication



Using hash code to provide message authentication



Only the hash code is encrypted using symmetric encryption.

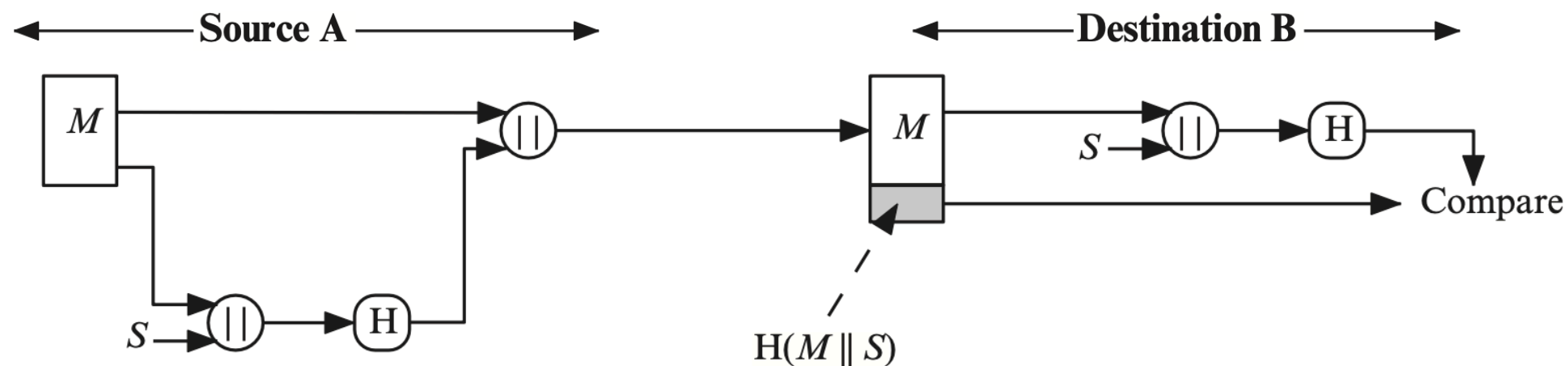
- This reduces the processing burden.
- It does not provide confidentiality.

Cryptographic Hash Functions

Message Authentication



Using hash code to provide message authentication



It uses a hash function for message authentication without encryption.

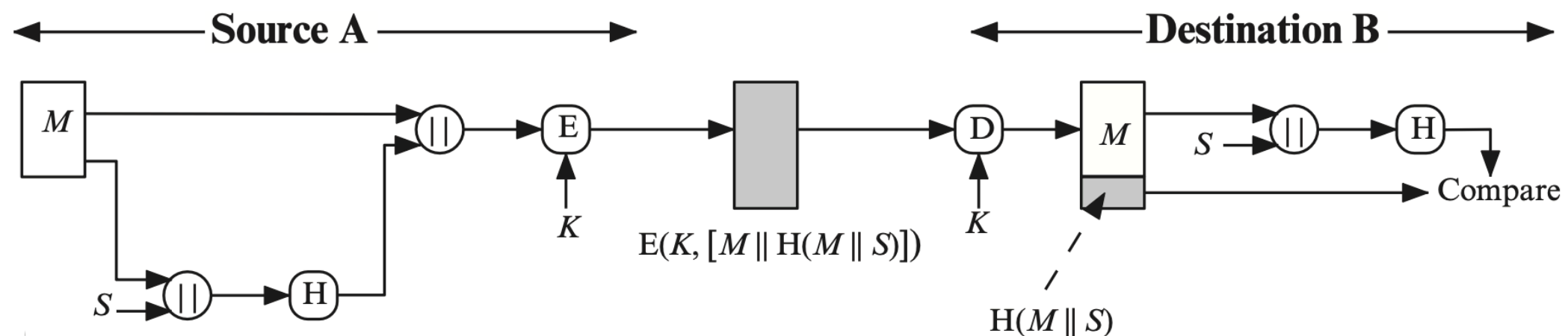
- It requires that two communicating parties share a common secret value S .
- The hash value is computed over the concatenation of M and S .

Cryptographic Hash Functions



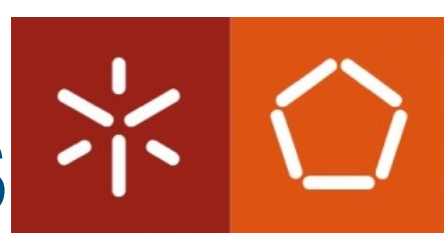
Message Authentication

Using hash code to provide message authentication



It adds confidentiality to the previous approach by encrypting the entire message plus the hash code.

Cryptographic Hash Functions



Requirements & Security

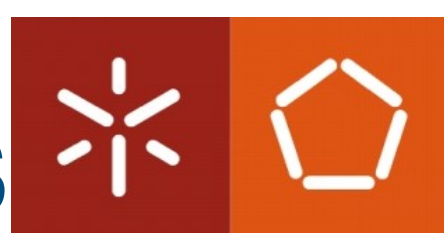
Terminology

For a hash value $h = H(x)$, x is called the **preimage** of h .

A **collision** occurs when $x \neq y$ and $H(x) = H(y)$.

- Since H is a many-to-one mapping, for any given h , there will be multiple possible preimages.
 - It maps a variable-length block (n bits) to a fixed-size value h (b bits). As typically $n > b$, on average, each h corresponds to 2^{n-b} .
- For data integrity, collisions are undesired.

Cryptographic Hash Functions

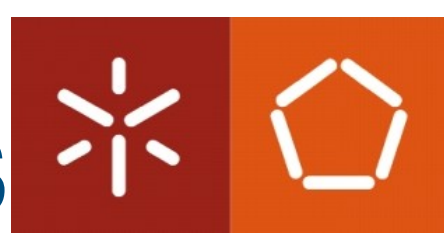


Requirements & Security

Security requirements for cryptographic use of hash functions

- » **Variable input size:** H can be applied to a block of data of any size;
- » **Fixed output size:** H produces a fixed-length output;
- » **Efficiency:** $H(x)$ is relatively easy to compute for any x (for software and hardware implementations);
- » **Preimage resistant:** For any given hash value h , it is computationally infeasible to find y such that $H(y) = h$;
- » **Second preimage resistant:** For any given block x , it is computationally infeasible to find $y \neq x$ with $H(y) = H(x)$;
- » **Collision resistant:** It is computationally infeasible to find any pair (x, y) with $x \neq y$, such that $H(x) = H(y)$;
- » **Pseudorandomness:** Output of H meets standard tests for randomness and unpredictability.

Cryptographic Hash Functions



Brute-Force Attacks

Preimage and Second Preimage Attacks

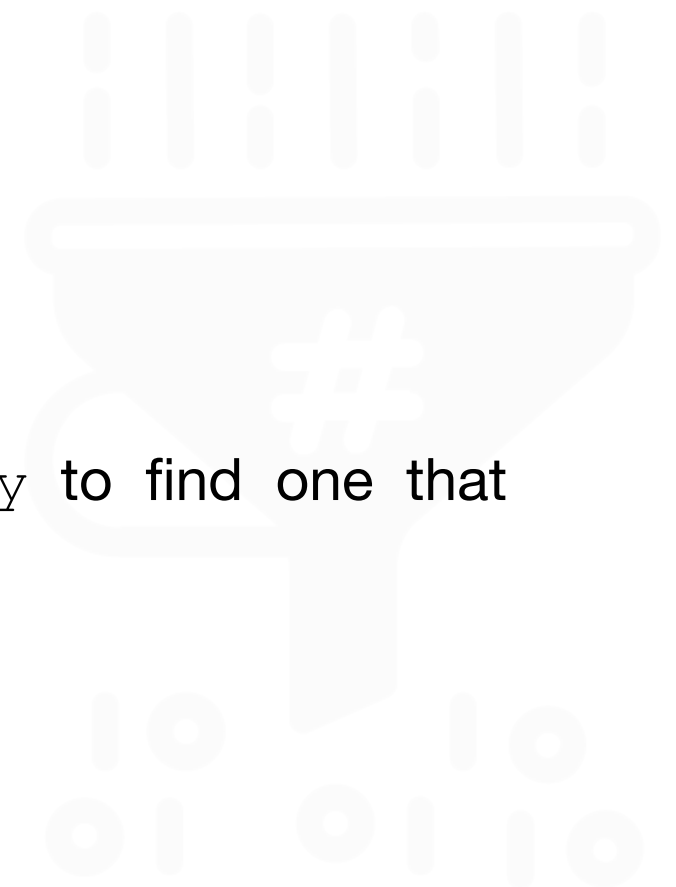
Objective

Find a value y such that $H(y)$ is equal to a given hash value h .

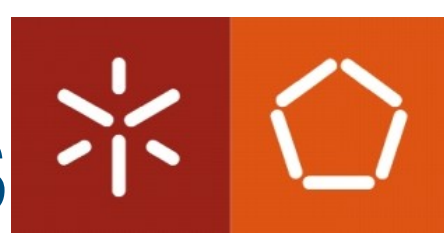
Brute-force method

Pick values of y at random and try each value until a collision occurs.

- For an m -bit hash value, the level of effort is proportional to 2^m ;
- On average, an adversary would have to try 2^{m-1} values of y to find one that generates a given hash value h .



Cryptographic Hash Functions



Brute-Force Attacks

Collision resistant attacks / Birthday attack

Objective

Find two messages or data blocks, x and y , that yield the same hash function:

$$H(x) = H(y)$$

Brute-force method

It relies on the birthday paradox in probability theory:

In a group of 23 people, there is a 50% chance that at least two of them have the same birthday. Increasing the group to 50 people, the probability is over 97%.

- When choosing random variables from a uniform distribution in the range 0 through $N - 1$, the probability that a repeated element is encountered exceeds 50% after $\sqrt{N}$ choices have been made.
- Thus, for an m -bit hash value, by picking blocks randomly, it is expected to find two blocks with the same hash value within $2^{m/2}$ attempts.
 - For 128-bit and 512-bit hash functions, an attack should test 2^{64} and 2^{256} messages.

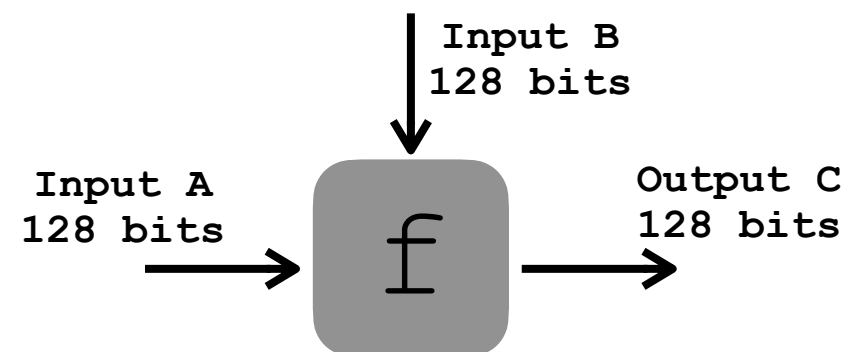
Cryptographic Hash Functions



Design

Compression function

A widely used component in secure hash function design. It consists of taking two inputs and producing an n -bit output.



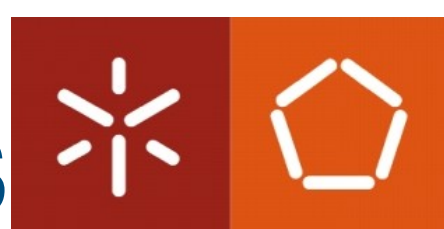
They are one-way functions that:

- Having the two inputs, it is easy to compute the output;
- It is infeasible to discover any of the inputs from the output;
- Having the output and one of the inputs, it is infeasible to find the second input;
- They are collision-resistant.



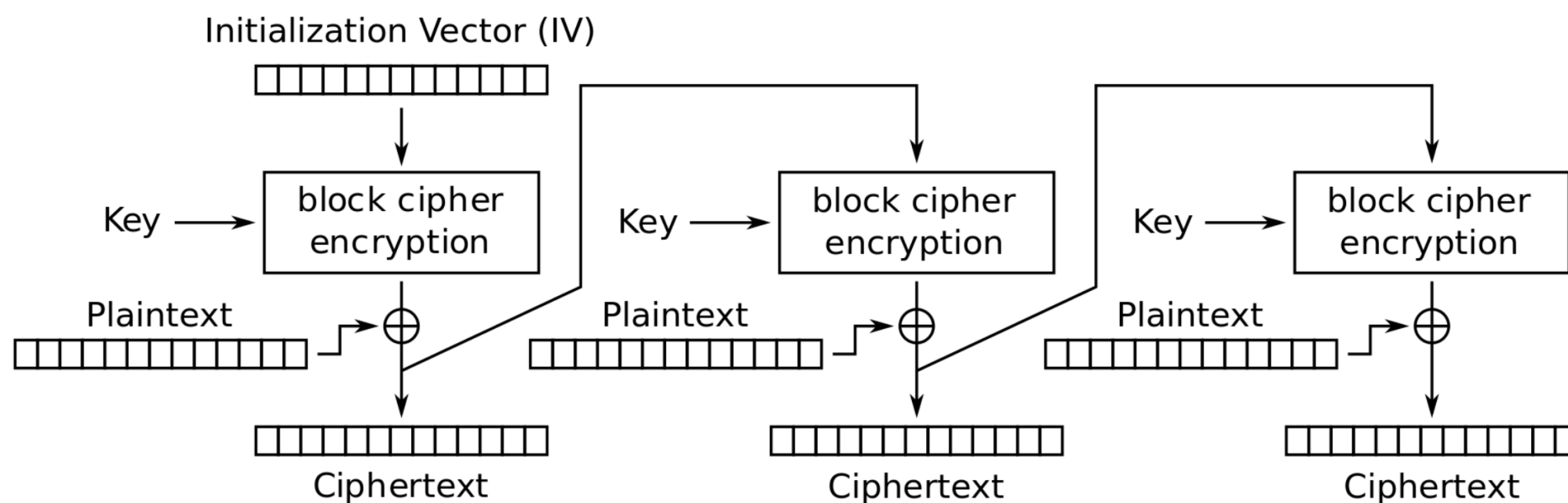
Cryptographic Hash Functions

Design



Compression function

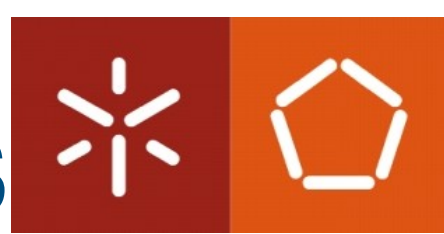
Can we use chained block ciphers as hash functions?



Cipher Feedback (CFB)

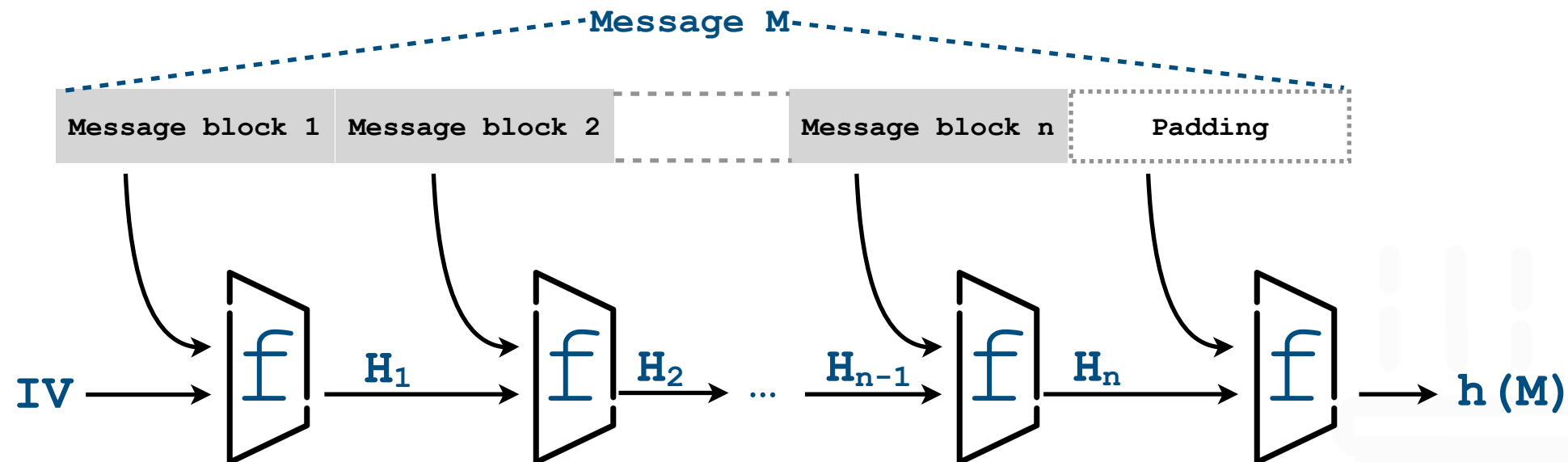
Cryptographic Hash Functions

Design



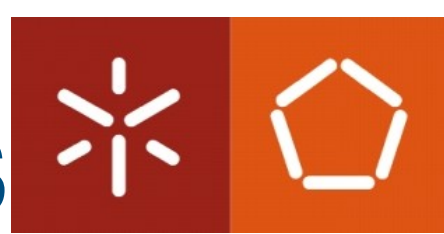
Merkle-Damgård structure

Used by most of the current cryptographic hash functions, including **SHA**.



- The compression function chaining propagates the previous message hash;
- The IV is a fixed value defined by the algorithm;
- The padding should include information about the message size;
- If the compression function f is collision-resistant, so is the hash function.

Cryptographic Hash Functions



Examples

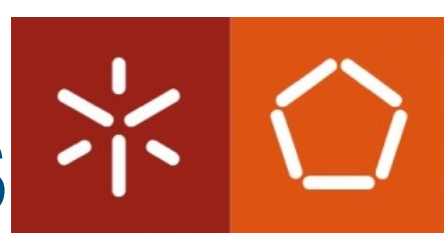
Not recommended for applications with high-security requirements due to collisions already encountered.

Algorithm	Block size (bits)	Rounds	Digest size (bits)
MD5	512	4	128
SHA-1	512	4	160
SHA-224	512	64	224
SHA-256	1024	64	256
SHA-384		80	384
SHA-512			512
SHA3-224	1152	24	224
SHA3-256	1088		256
SHA3-384	832		384
SHA3-512	576		512

- SHA-2 Standard
- Considered secure for cryptographic application

- New standard from 2015;
- Uses Keccak's algorithm;
- Based on the sponge-construction with a wide range of configuration parameters that allow to adjust the level of security/efficiency.

Cryptographic Hash Functions



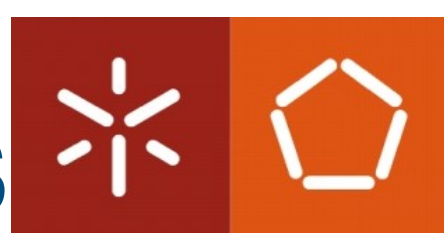
Hands-on

The passwords of an online service have been leaked on the internet. It has been noted that the password management in this service uses SHA256 and stores the passwords in hexadecimal format. In other words, the user's password is stored as follows: `hex(SHA256(password))`.

Knowing the code below represents a password from this leak and that it is also part of the top 10k most common passwords, create a program that identifies the password. Which technique did you use?

```
b23a6a8439c0dde5515893e7c90c1e3233b8616e634470f20dc4928bcf3609bc
```

Cryptographic Hash Functions

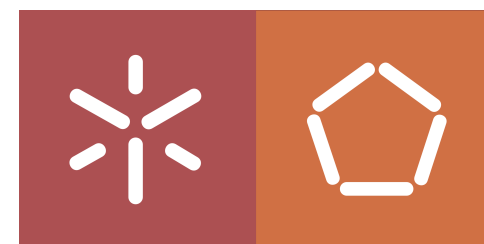


Hands-on

Extend the code from the previous exercise to include a mechanism to guarantee the integrity of the transmitted data.

Suggestion:

- Use hash functions available in Python's Cryptography library;
- Send the data and the hash value within the same structure, e.g., a dictionary as a *JSON (JavaScript Object Notation)* object.



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